

8.5 HCS08 Instruction Set Summary

Instruction Set Summary Nomenclature

The nomenclature listed here is used in the instruction descriptions in [Table 8-2](#).

Operators

()	=	Contents of register or memory location shown inside parentheses
←	=	Is loaded with (read: “gets”)
&	=	Boolean AND
	=	Boolean OR
⊕	=	Boolean exclusive-OR
×	=	Multiply
÷	=	Divide
:	=	Concatenate
+	=	Add
–	=	Negate (two’s complement)

CPU registers

A	=	Accumulator
CCR	=	Condition code register
H	=	Index register, higher order (most significant) 8 bits
X	=	Index register, lower order (least significant) 8 bits
PC	=	Program counter
PCH	=	Program counter, higher order (most significant) 8 bits
PCL	=	Program counter, lower order (least significant) 8 bits
SP	=	Stack pointer

Memory and addressing

M	=	A memory location or absolute data, depending on addressing mode
M:M + 0x0001	=	A 16-bit value in two consecutive memory locations. The higher-order (most significant) 8 bits are located at the address of M, and the lower-order (least significant) 8 bits are located at the next higher sequential address.

Condition code register (CCR) bits

V	=	Two’s complement overflow indicator, bit 7
H	=	Half carry, bit 4
I	=	Interrupt mask, bit 3
N	=	Negative indicator, bit 2
Z	=	Zero indicator, bit 1
C	=	Carry/borrow, bit 0 (carry out of bit 7)

CCR activity notation

–	=	Bit not affected
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0	=	Bit forced to 0
1	=	Bit forced to 1
↑	=	Bit set or cleared according to results of operation
U	=	Undefined after the operation

Machine coding notation

dd	=	Low-order 8 bits of a direct address 0x0000–0x00FF (high byte assumed to be 0x00)
ee	=	Upper 8 bits of 16-bit offset
ff	=	Lower 8 bits of 16-bit offset or 8-bit offset
ii	=	One byte of immediate data
jj	=	High-order byte of a 16-bit immediate data value
kk	=	Low-order byte of a 16-bit immediate data value
hh	=	High-order byte of 16-bit extended address
ll	=	Low-order byte of 16-bit extended address
pg	=	Page
rr	=	Relative offset

Source form

Everything in the source forms columns, *except expressions in italic characters*, is literal information that must appear in the assembly source file exactly as shown. The initial 3- to 5-letter mnemonic is always a literal expression. All commas, pound signs (#), parentheses, and plus signs (+) are literal characters.

<i>n</i>	—	Any label or expression that evaluates to a single integer in the range 0–7
<i>opr8i</i>	—	Any label or expression that evaluates to an 8-bit immediate value
<i>opr16i</i>	—	Any label or expression that evaluates to a 16-bit immediate value
<i>opr8a</i>	—	Any label or expression that evaluates to an 8-bit value. The instruction treats this 8-bit value as the low order 8 bits of an address in the direct page of the 64-Kbyte address space (0x00xx).
<i>opr16a</i>	—	Any label or expression that evaluates to a 16-bit value. The instruction treats this value as an address in the 64-Kbyte address space.
<i>opr8</i>	—	Any label or expression that evaluates to an unsigned 8-bit value, used for indexed addressing
<i>opr16</i>	—	Any label or expression that evaluates to a 16-bit value. Because the HCS08 has a 16-bit address bus, this can be either a signed or an unsigned value.
<i>page</i>	—	Any label or expression that evaluates to a valid bank number for the PPAGE register. For a 128-Kbyte derivative, any value between 0 and 7 is valid.
<i>rel</i>	—	Any label or expression that refers to an address that is within –128 to +127 locations from the next address after the last byte of object code for the current instruction. The assembler will calculate the 8-bit signed offset and include it in the object code for this instruction.

Address modes

INH	=	Inherent (no operands)
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IMM	=	8-bit or 16-bit immediate
DIR	=	8-bit direct
EXT	=	16-bit extended
IX	=	16-bit indexed no offset
IX+	=	16-bit indexed no offset, post increment (CBEQ and MOV only)
IX1	=	16-bit indexed with 8-bit offset from H:X
IX1+	=	16-bit indexed with 8-bit offset, post increment (CBEQ only)
IX2	=	16-bit indexed with 16-bit offset from H:X
REL	=	8-bit relative offset
SP1	=	Stack pointer with 8-bit offset
SP2	=	Stack pointer with 16-bit offset

Table 8-2. HCS08 Instruction Set Summary (Sheet 1 of 7)

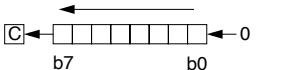
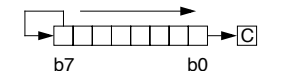
Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles ¹
			V	H	I	N	Z	C				
ADC #opr8i ADC opr8a ADC opr16a ADC oprx16,X ADC oprx8,X ADC ,X ADC oprx16,SP ADC oprx8,SP	Add with Carry	$A \leftarrow (A) + (M) + (C)$	↑	↑	—	↑	↑	↑	IMM DIR EXT IX2 IX1 IX SP2 SP1	A9 ii B9 dd C9 hh ll D9 ee ff E9 ff F9 ff 9ED9 ee ff 9EE9 ff		2 3 4 4 3 3 5 4
ADD #opr8i ADD opr8a ADD opr16a ADD oprx16,X ADD oprx8,X ADD ,X ADD oprx16,SP ADD oprx8,SP	Add without Carry	$A \leftarrow (A) + (M)$	↑	↑	—	↑	↑	↑	IMM DIR EXT IX2 IX1 IX SP2 SP1	AB ii BB dd CB hh ll DB ee ff EB ff FB ff 9EDB ee ff 9EEB ff		2 3 4 4 3 3 5 4
AIS #opr8i	Add Immediate Value (Signed) to Stack Pointer	$SP \leftarrow (SP) + (M)$ M is sign extended to a 16-bit value	—	—	—	—	—	—	IMM	A7 ii		2
AIX #opr8i	Add Immediate Value (Signed) to Index Register (H:X)	$H:X \leftarrow (H:X) + (M)$ M is sign extended to a 16-bit value	—	—	—	—	—	—	IMM	AF ii		2
AND #opr8i AND opr8a AND opr16a AND oprx16,X AND oprx8,X AND ,X AND oprx16,SP AND oprx8,SP	Logical AND	$A \leftarrow (A) \& (M)$	0	—	—	↑	↑	—	IMM DIR EXT IX2 IX1 IX SP2 SP1	A4 ii B4 dd C4 hh ll D4 ee ff E4 ff F4 ff 9ED4 ee ff 9EE4 ff		2 3 4 4 3 3 5 4
ASL opr8a ASLA ASLX ASL oprx8,X ASL ,X ASL oprx8,SP	Arithmetic Shift Left (Same as LSL)		↑	—	—	↑	↑	↑	DIR INH INH IX1 IX SP1	38 dd 48 dd 58 ff 68 ff 78 ff 9E68 ff		5 1 1 5 4 6
ASR opr8a ASRA ASRX ASR oprx8,X ASR ,X ASR oprx8,SP	Arithmetic Shift Right		↑	—	—	↑	↑	↑	DIR INH INH IX1 IX SP1	37 dd 47 dd 57 ff 67 ff 77 ff 9E67 ff		5 1 1 5 4 6
BCC rel	Branch if Carry Bit Clear	Branch if (C) = 0	—	—	—	—	—	—	REL	24 rr		3

Table 8-2. HCS08 Instruction Set Summary (Sheet 2 of 7)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles ¹
			V	H	I	N	Z	C				
BCLR <i>n,opr8a</i>	Clear Bit n in Memory	$M_n \leftarrow 0$	–	–	–	–	–	–	DIR (b0) DIR (b1) DIR (b2) DIR (b3) DIR (b4) DIR (b5) DIR (b6) DIR (b7)	11 13 15 17 19 1B 1D 1F	dd dd dd dd dd dd dd dd	5 5 5 5 5 5 5 5
BCS <i>rel</i>	Branch if Carry Bit Set (Same as BLO)	Branch if (C) = 1	–	–	–	–	–	–	REL	25	rr	3
BEQ <i>rel</i>	Branch if Equal	Branch if (Z) = 1	–	–	–	–	–	–	REL	27	rr	3
BGE <i>rel</i>	Branch if Greater Than or Equal To (Signed Operands)	Branch if $(N \oplus V) = 0$	–	–	–	–	–	–	REL	90	rr	3
BGND	Enter Active Background if ENBDM = 1	Waits For and Processes BDM Commands Until GO, TRACE1, or TAGGO	–	–	–	–	–	–	INH	82		5+
BGT <i>rel</i>	Branch if Greater Than (Signed Operands)	Branch if $(Z) \mid (N \oplus V) = 0$	–	–	–	–	–	–	REL	92	rr	3
BHCC <i>rel</i>	Branch if Half Carry Bit Clear	Branch if (H) = 0	–	–	–	–	–	–	REL	28	rr	3
BHCS <i>rel</i>	Branch if Half Carry Bit Set	Branch if (H) = 1	–	–	–	–	–	–	REL	29	rr	3
BHI <i>rel</i>	Branch if Higher	Branch if $(C) \mid (Z) = 0$	–	–	–	–	–	–	REL	22	rr	3
BHS <i>rel</i>	Branch if Higher or Same (Same as BCC)	Branch if (C) = 0	–	–	–	–	–	–	REL	24	rr	3
BIH <i>rel</i>	Branch if IRQ Pin High	Branch if IRQ pin = 1	–	–	–	–	–	–	REL	2F	rr	3
BIL <i>rel</i>	Branch if IRQ Pin Low	Branch if IRQ pin = 0	–	–	–	–	–	–	REL	2E	rr	3
BIT <i>#opr8i</i> BIT <i>opr8a</i> BIT <i>opr16a</i> BIT <i>opr16,X</i> BIT <i>opr8,X</i> BIT <i>,X</i> BIT <i>opr16,SP</i> BIT <i>opr8,SP</i>	Bit Test	(A) & (M) (CCR Updated but Operands Not Changed)	0	–	–	↑	↑	–	IMM DIR EXT IX2 IX1 IX SP2 SP1	A5 B5 C5 D5 E5 F5 9ED5 9EE5	ii dd hh ll ee ff ff ee ff ff	2 3 4 4 3 3 5 4
BLE <i>rel</i>	Branch if Less Than or Equal To (Signed Operands)	Branch if $(Z) \mid (N \oplus V) = 1$	–	–	–	–	–	–	REL	93	rr	3
BLO <i>rel</i>	Branch if Lower (Same as BCS)	Branch if (C) = 1	–	–	–	–	–	–	REL	25	rr	3
BLS <i>rel</i>	Branch if Lower or Same	Branch if $(C) \mid (Z) = 1$	–	–	–	–	–	–	REL	23	rr	3
BLT <i>rel</i>	Branch if Less Than (Signed Operands)	Branch if $(N \oplus V) = 1$	–	–	–	–	–	–	REL	91	rr	3
BMC <i>rel</i>	Branch if Interrupt Mask Clear	Branch if (I) = 0	–	–	–	–	–	–	REL	2C	rr	3
BMI <i>rel</i>	Branch if Minus	Branch if (N) = 1	–	–	–	–	–	–	REL	2B	rr	3
BMS <i>rel</i>	Branch if Interrupt Mask Set	Branch if (I) = 1	–	–	–	–	–	–	REL	2D	rr	3
BNE <i>rel</i>	Branch if Not Equal	Branch if (Z) = 0	–	–	–	–	–	–	REL	26	rr	3
BPL <i>rel</i>	Branch if Plus	Branch if (N) = 0	–	–	–	–	–	–	REL	2A	rr	3
BRA <i>rel</i>	Branch Always	No Test	–	–	–	–	–	–	REL	20	rr	3

Table 8-2. HCS08 Instruction Set Summary (Sheet 3 of 7)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles ¹
			V	H	I	N	Z	C				
BRCLR <i>n,opr8a,rel</i>	Branch if Bit <i>n</i> in Memory Clear	Branch if (Mn) = 0	–	–	–	–	–	↑	DIR (b0) DIR (b1) DIR (b2) DIR (b3) DIR (b4) DIR (b5) DIR (b6) DIR (b7)	01 03 05 07 09 0B 0D 0F	dd rr dd rr dd rr dd rr dd rr dd rr dd rr dd rr	5 5 5 5 5 5 5 5
BRN <i>rel</i>	Branch Never	Uses 3 Bus Cycles	–	–	–	–	–	–	REL	21	rr	3
BRSET <i>n,opr8a,rel</i>	Branch if Bit <i>n</i> in Memory Set	Branch if (Mn) = 1	–	–	–	–	–	↑	DIR (b0) DIR (b1) DIR (b2) DIR (b3) DIR (b4) DIR (b5) DIR (b6) DIR (b7)	00 02 04 06 08 0A 0C 0E	dd rr dd rr dd rr dd rr dd rr dd rr dd rr dd rr	5 5 5 5 5 5 5 5
BSET <i>n,opr8a</i>	Set Bit <i>n</i> in Memory	Mn ← 1	–	–	–	–	–	–	DIR (b0) DIR (b1) DIR (b2) DIR (b3) DIR (b4) DIR (b5) DIR (b6) DIR (b7)	10 12 14 16 18 1A 1C 1E	dd dd dd dd dd dd dd dd	5 5 5 5 5 5 5 5
BSR <i>rel</i>	Branch to Subroutine	PC ← (PC) + 0x0002 push (PCL); SP ← (SP) – 0x0001 push (PCH); SP ← (SP) – 0x0001 PC ← (PC) + <i>rel</i>	–	–	–	–	–	–	REL	AD	rr	5
CALL page, opr16a	Call Subroutine	PC ← PC + 4 Push (PCL); SP ← (SP) – 0x0001 Push (PCH); SP ← (SP) – 0x0001 Push (PPAGE); SP ← (SP) – 0x0001 PPAGE ← page PC ← Unconditional Address	–	–	–	–	–	–	EXT	AC	pghll	8
CBEQ <i>opr8a,rel</i> CBEQA # <i>opr8i,rel</i> CBEQX # <i>opr8i,rel</i> CBEQ <i>opr8,X+,rel</i> CBEQ <i>,X+,rel</i> CBEQ <i>opr8,SP,rel</i>	Compare and Branch if Equal	Branch if (A) = (M) Branch if (A) = (M) Branch if (X) = (M) Branch if (A) = (M) Branch if (A) = (M) Branch if (A) = (M)	–	–	–	–	–	–	DIR IMM IMM IX1+ IX+ SP1	31 41 51 61 71 9E61	dd rr ii rr ii rr ff rr rr rr ff rr	5 4 4 5 5 6
CLC	Clear Carry Bit	C ← 0	–	–	–	–	–	0	INH	98		1
CLI	Clear Interrupt Mask Bit	I ← 0	–	–	0	–	–	–	INH	9A		1
CLR <i>opr8a</i> CLRA CLR X CLR H CLR <i>opr8,X</i> CLR <i>,X</i> CLR <i>opr8,SP</i>	Clear	M ← 0x00 A ← 0x00 X ← 0x00 H ← 0x00 M ← 0x00 M ← 0x00 M ← 0x00	0	–	–	0	1	–	DIR INH INH INH IX1 IX SP1	3F 4F 5F 8C 6F 7F 9E6F	dd ff ff ff	5 1 1 1 5 4 6
CMP # <i>opr8i</i> CMP <i>opr8a</i> CMP <i>opr16a</i> CMP <i>opr16,X</i> CMP <i>opr8,X</i> CMP <i>,X</i> CMP <i>opr16,SP</i> CMP <i>opr8,SP</i>	Compare Accumulator with Memory	(A) – (M) (CCR Updated But Operands Not Changed)	↑	–	–	↑	↑	↑	IMM DIR EXT IX2 IX1 IX SP2 SP1	A1 B1 C1 D1 E1 F1 9ED1 9EE1	ii dd hh ll ee ff ff ee ff ff	2 3 4 4 3 3 5 4

Table 8-2. HCS08 Instruction Set Summary (Sheet 4 of 7)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles ¹
			V	H	I	N	Z	C				
COM <i>opr8a</i> COMA COMX COM <i>opr8,X</i> COM <i>,X</i> COM <i>opr8,SP</i>	Complement (One's Complement)	$M \leftarrow (\overline{M}) = 0xFF - (M)$ $A \leftarrow (\overline{A}) = 0xFF - (A)$ $X \leftarrow (\overline{X}) = 0xFF - (X)$ $M \leftarrow (\overline{M}) = 0xFF - (M)$ $M \leftarrow (\overline{M}) = 0xFF - (M)$ $M \leftarrow (\overline{M}) = 0xFF - (M)$	0	–	–	↑	↑	1	DIR INH INH IX1 IX SP1	33 43 53 63 73 9E63	dd ff ff	5 1 1 5 4 6
CPHX <i>opr16a</i> CPHX <i>#opr16i</i> CPHX <i>opr8a</i> CPHX <i>opr8,SP</i>	Compare Index Register (H:X) with Memory	(H:X) – (M:M + 0x0001) (CCR Updated But Operands Not Changed)	↑	–	–	↑	↑	↑	EXT IMM DIR SP1	3E 65 75 9EF3	hh ll jj kk dd ff	6 3 5 6
CPX <i>#opr8i</i> CPX <i>opr8a</i> CPX <i>opr16a</i> CPX <i>opr8,X</i> CPX <i>opr8,X</i> CPX <i>,X</i> CPX <i>opr8,SP</i> CPX <i>opr8,SP</i>	Compare X (Index Register Low) with Memory	(X) – (M) (CCR Updated But Operands Not Changed)	↑	–	–	↑	↑	↑	IMM DIR EXT IX2 IX1 IX SP2 SP1	A3 B3 C3 D3 E3 F3 9ED3 9EE3	ii dd hh ll ee ff ff ff ff	2 3 4 4 3 3 5 4
DAA	Decimal Adjust Accumulator After ADD or ADC of BCD Values	(A) ₁₀	U	–	–	↑	↑	↑	INH	72		1
DBNZ <i>opr8a,rel</i> DBNZ <i>rel</i> DBNZX <i>rel</i> DBNZ <i>opr8,X,rel</i> DBNZ <i>,X,rel</i> DBNZ <i>opr8,SP,rel</i>	Decrement and Branch if Not Zero	Decrement A, X, or M Branch if (result) ≠ 0 DBNZX Affects X Not H	–	–	–	–	–	–	DIR INH INH IX1 IX SP1	3B 4B 5B 6B 7B 9E6B	dd rr rr rr ff rr rr rr	7 4 4 7 6 8
DEC <i>opr8a</i> DECA DECX DEC <i>opr8,X</i> DEC <i>,X</i> DEC <i>opr8,SP</i>	Decrement	$M \leftarrow (M) - 0x01$ $A \leftarrow (A) - 0x01$ $X \leftarrow (X) - 0x01$ $M \leftarrow (M) - 0x01$ $M \leftarrow (M) - 0x01$ $M \leftarrow (M) - 0x01$	↑	–	–	↑	↑	–	DIR INH INH IX1 IX SP1	3A 4A 5A 6A 7A 9E6A	dd ff ff	5 1 1 5 4 6
DIV	Divide	$A \leftarrow (H:A) \div (X)$ H ← Remainder	–	–	–	–	↑	↑	INH	52		6
EOR <i>#opr8i</i> EOR <i>opr8a</i> EOR <i>opr16a</i> EOR <i>opr8,X</i> EOR <i>opr8,X</i> EOR <i>,X</i> EOR <i>opr8,SP</i> EOR <i>opr8,SP</i>	Exclusive OR Memory with Accumulator	$A \leftarrow (A \oplus M)$	0	–	–	↑	↑	–	IMM DIR EXT IX2 IX1 IX SP2 SP1	A8 B8 C8 D8 E8 F8 9ED8 9EE8	ii dd hh ll ee ff ff ff ff	2 3 4 4 3 3 5 4
INC <i>opr8a</i> INCA INCX INC <i>opr8,X</i> INC <i>,X</i> INC <i>opr8,SP</i>	Increment	$M \leftarrow (M) + 0x01$ $A \leftarrow (A) + 0x01$ $X \leftarrow (X) + 0x01$ $M \leftarrow (M) + 0x01$ $M \leftarrow (M) + 0x01$ $M \leftarrow (M) + 0x01$	↑	–	–	↑	↑	–	DIR INH INH IX1 IX SP1	3C 4C 5C 6C 7C 9E6C	dd ff ff	5 1 1 5 4 6
JMP <i>opr8a</i> JMP <i>opr16a</i> JMP <i>opr8,X</i> JMP <i>opr8,X</i> JMP <i>,X</i>	Jump	PC ← Jump Address	–	–	–	–	–	–	DIR EXT IX2 IX1 IX	BC CC DC EC FC	dd ll hh ll ee ff ff	3 4 4 3 3
JSR <i>opr8a</i> JSR <i>opr16a</i> JSR <i>opr8,X</i> JSR <i>opr8,X</i> JSR <i>,X</i>	Jump to Subroutine	PC ← (PC) + n (n = 1, 2, or 3) Push (PCL); SP ← (SP) – 0x0001 Push (PCH); SP ← (SP) – 0x0001 PC ← Unconditional Address	–	–	–	–	–	–	DIR EXT IX2 IX1 IX	BD CD DD ED FD	dd ll hh ll ee ff ff	5 6 6 5 5

Table 8-2. HCS08 Instruction Set Summary (Sheet 5 of 7)

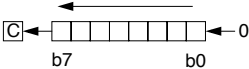
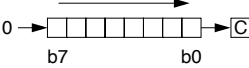
Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles ¹
			V	H	I	N	Z	C				
LDA #opr8i LDA opr8a LDA opr16a LDA oprx16,X LDA oprx8,X LDA ,X LDA oprx16,SP LDA oprx8,SP	Load Accumulator from Memory	$A \leftarrow (M)$	0	–	–	↑	↑	–	IMM DIR EXT IX2 IX1 IX SP2 SP1	A6 B6 C6 D6 E6 F6 9ED6 9EE6	ii dd hh ll ee ff ff ee ff ff	2 3 4 4 3 3 5 4
LDHX #opr16i LDHX opr8a LDHX opr16a LDHX ,X LDHX oprx16,X LDHX oprx8,X LDHX oprx8,SP	Load Index Register (H:X) from Memory	$H:X \leftarrow (M:M + 0x0001)$	0	–	–	↑	↑	–	IMM DIR EXT IX IX2 IX1 SP1	45 55 32 9EAE 9EBE 9ECE 9EFE	jj kk dd ll hh ll ee ff ff ff	3 4 5 5 6 5 5
LDX #opr8i LDX opr8a LDX opr16a LDX oprx16,X LDX oprx8,X LDX ,X LDX oprx16,SP LDX oprx8,SP	Load X (Index Register Low) from Memory	$X \leftarrow (M)$	0	–	–	↑	↑	–	IMM DIR EXT IX2 IX1 IX SP2 SP1	AE BE CE DE EE FE 9EDE 9EEE	ii dd hh ll ee ff ff ee ff ff	2 3 4 4 3 3 5 4
LSL opr8a LSLA LSLX LSL oprx8,X LSL ,X LSL oprx8,SP	Logical Shift Left (Same as ASL)		↑	–	–	↑	↑	↑	DIR INH INH IX1 IX SP1	38 48 58 68 78 9E78	dd ff ff	5 1 1 5 4 6
LSR opr8a LSRA LSRX LSR oprx8,X LSR ,X LSR oprx8,SP	Logical Shift Right		↑	–	–	0	↑	↑	DIR INH INH IX1 IX SP1	34 44 54 64 74 9E64	dd ff ff	5 1 1 5 4 6
MOV opr8a,opr8a MOV opr8a,X+ MOV #opr8i,opr8a MOV ,X+,opr8a	Move	$(M)_{\text{destination}} \leftarrow (M)_{\text{source}}$ $H:X \leftarrow (H:X) + 0x0001$ in IX+/DIR and DIR/IX+ Modes	0	–	–	↑	↑	–	DIR/DIR DIR/IX+ IMM/DIR IX+/DIR	4E 5E 6E 7E	dd dd dd dd ii dd dd	5 5 4 5
MUL	Unsigned multiply	$X:A \leftarrow (X) \times (A)$	–	0	–	–	–	0	INH	42		5
NEG opr8a NEGA NEGX NEG oprx8,X NEG ,X NEG oprx8,SP	Negate (Two's Complement)	$M \leftarrow -(M) = 0x00 - (M)$ $A \leftarrow -(A) = 0x00 - (A)$ $X \leftarrow -(X) = 0x00 - (X)$ $M \leftarrow -(M) = 0x00 - (M)$ $M \leftarrow -(M) = 0x00 - (M)$ $M \leftarrow -(M) = 0x00 - (M)$		–	–	↑	↑	↑	DIR INH INH IX1 IX SP1	30 40 50 60 70 9E60	dd ff ff	5 1 1 5 4 6
NOP	No Operation	Uses 1 Bus Cycle	–	–	–	–	–	–	INH	9D		1
NSA	Nibble Swap Accumulator	$A \leftarrow (A[3:0]:A[7:4])$	–	–	–	–	–	–	INH	62		1
ORA #opr8i ORA opr8a ORA opr16a ORA oprx16,X ORA oprx8,X ORA ,X ORA oprx16,SP ORA oprx8,SP	Inclusive OR Accumulator and Memory	$A \leftarrow (A) \text{ I } (M)$	0	–	–	↑	↑	–	IMM DIR EXT IX2 IX1 IX SP2 SP1	AA BA CA DA EA FA 9EDA 9EEA	ii dd hh ll ee ff ff ee ff ff	2 3 4 4 3 3 5 4
PSHA	Push Accumulator onto Stack	Push (A); $SP \leftarrow (SP) - 0x0001$	–	–	–	–	–	–	INH	87		2
PSHH	Push H (Index Register High) onto Stack	Push (H); $SP \leftarrow (SP) - 0x0001$	–	–	–	–	–	–	INH	8B		2
PSHX	Push X (Index Register Low) onto Stack	Push (X); $SP \leftarrow (SP) - 0x0001$	–	–	–	–	–	–	INH	89		2

Table 8-2. HCS08 Instruction Set Summary (Sheet 6 of 7)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles ¹
			V	H	I	N	Z	C				
PULA	Pull Accumulator from Stack	$SP \leftarrow (SP + 0x0001); \text{Pull (A)}$	–	–	–	–	–	–	INH	86		3
PULH	Pull H (Index Register High) from Stack	$SP \leftarrow (SP + 0x0001); \text{Pull (H)}$	–	–	–	–	–	–	INH	8A		3
PULX	Pull X (Index Register Low) from Stack	$SP \leftarrow (SP + 0x0001); \text{Pull (X)}$	–	–	–	–	–	–	INH	88		3
ROL <i>opr8a</i> ROLA ROLX ROL <i>opr8,X</i> ROL <i>,X</i> ROL <i>opr8,SP</i>	Rotate Left through Carry		↑	–	–	↑	↑	↑	DIR INH INH IX1 IX SP1	39 49 59 69 79 9E69	dd ff ff	5 1 1 5 4 6
ROR <i>opr8a</i> RORA RORX ROR <i>opr8,X</i> ROR <i>,X</i> ROR <i>opr8,SP</i>	Rotate Right through Carry		↑	–	–	↑	↑	↑	DIR INH INH IX1 IX SP1	36 46 56 66 76 9E66	dd ff ff	5 1 1 5 4 6
RSP	Reset Stack Pointer	$SP \leftarrow 0xFF$ (High Byte Not Affected)	–	–	–	–	–	–	INH	9C		1
RTC	Return from CALL	$SP \leftarrow (SP) + 0x0001; \text{Pull (PPAGE)}$ $SP \leftarrow (SP) + 0x0001; \text{Pull (PCH)}$ $SP \leftarrow (SP) + 0x0001; \text{Pull (PCL)}$	–	–	–	–	–	–	INH	8D		7
RTI	Return from Interrupt	$SP \leftarrow (SP) + 0x0001; \text{Pull (CCR)}$ $SP \leftarrow (SP) + 0x0001; \text{Pull (A)}$ $SP \leftarrow (SP) + 0x0001; \text{Pull (X)}$ $SP \leftarrow (SP) + 0x0001; \text{Pull (PCH)}$ $SP \leftarrow (SP) + 0x0001; \text{Pull (PCL)}$	↑	↑	↑	↑	↑	↑	INH	80		9
RTS	Return from Subroutine	$SP \leftarrow SP + 0x0001; \text{Pull (PCH)}$ $SP \leftarrow SP + 0x0001; \text{Pull (PCL)}$	–	–	–	–	–	–	INH	81		6
SBC <i>#opr8i</i> SBC <i>opr8a</i> SBC <i>opr16a</i> SBC <i>opr16,X</i> SBC <i>opr8,X</i> SBC <i>,X</i> SBC <i>opr16,SP</i> SBC <i>opr8,SP</i>	Subtract with Carry	$A \leftarrow (A) - (M) - (C)$	↑	–	–	↑	↑	↑	IMM DIR EXT IX2 IX1 IX SP2 SP1	A2 B2 C2 D2 E2 F2 9ED2 9EE2	ii dd hh ll ee ff ee ff ee ff ee ff	2 3 4 4 3 3 5 4
SEC	Set Carry Bit	$C \leftarrow 1$	–	–	–	–	–	1	INH	99		1
SEI	Set Interrupt Mask Bit	$I \leftarrow 1$	–	–	1	–	–	–	INH	9B		1
STA <i>opr8a</i> STA <i>opr16a</i> STA <i>opr16,X</i> STA <i>opr8,X</i> STA <i>,X</i> STA <i>opr16,SP</i> STA <i>opr8,SP</i>	Store Accumulator in Memory	$M \leftarrow (A)$	0	–	–	↑	↑	–	DIR EXT IX2 IX1 IX SP2 SP1	B7 C7 D7 E7 F7 9ED7 9EE7	dd hh ll ee ff ff ff ee ff ff	3 4 4 3 2 5 4
STHX <i>opr8a</i> STHX <i>opr16a</i> STHX <i>opr8,SP</i>	Store H:X (Index Reg.)	$(M:M + 0x0001) \leftarrow (H:X)$	0	–	–	↑	↑	–	DIR EXT SP1	35 96 9EFF	dd hh ll ff	4 5 5
STOP	Enable Interrupts: Stop Processing Refer to MCU Documentation	$I \text{ bit} \leftarrow 0; \text{Stop Processing}$	–	–	0	–	–	–	INH	8E		2+
STX <i>opr8a</i> STX <i>opr16a</i> STX <i>opr16,X</i> STX <i>opr8,X</i> STX <i>,X</i> STX <i>opr16,SP</i> STX <i>opr8,SP</i>	Store X (Low 8 Bits of Index Register) in Memory	$M \leftarrow (X)$	0	–	–	↑	↑	–	DIR EXT IX2 IX1 IX SP2 SP1	BF CF DF EF FF 9EDF 9EEF	dd hh ll ee ff ff ff ee ff ff	3 4 4 3 2 5 4

Table 8-2. HCS08 Instruction Set Summary (Sheet 7 of 7)

Source Form	Operation	Description	Effect on CCR						Address Mode	Opcode	Operand	Bus Cycles ¹
			V	H	I	N	Z	C				
SUB #opr8i SUB opr8a SUB opr16a SUB oprx16,X SUB oprx8,X SUB ,X SUB oprx16,SP SUB oprx8,SP	Subtract	$A \leftarrow (A) - (M)$	↑	—	—	↑	↑	↑	IMM DIR EXT IX2 IX1 IX SP2 SP1	A0 B0 C0 D0 E0 F0 9ED0 9EE0	ii dd hh ll ee ff ff ff	2 3 4 4 3 3 5 4
SWI	Software Interrupt	$PC \leftarrow (PC) + 0x0001$ Push (PCL); $SP \leftarrow (SP) - 0x0001$ Push (PCH); $SP \leftarrow (SP) - 0x0001$ Push (X); $SP \leftarrow (SP) - 0x0001$ Push (A); $SP \leftarrow (SP) - 0x0001$ Push (CCR); $SP \leftarrow (SP) - 0x0001$ $I \leftarrow 1$; PCH ← Interrupt Vector High Byte PCL ← Interrupt Vector Low Byte	—	—	1	—	—	—	INH	83		11
TAP	Transfer Accumulator to CCR	$CCR \leftarrow (A)$	↑	↑	↑	↑	↑	↑	INH	84		1
TAX	Transfer Accumulator to X (Index Register Low)	$X \leftarrow (A)$	—	—	—	—	—	—	INH	97		1
TPA	Transfer CCR to Accumulator	$A \leftarrow (CCR)$	—	—	—	—	—	—	INH	85		1
TST opr8a TSTA TSTX TST oprx8,X TST ,X TST oprx8,SP	Test for Negative or Zero	$(M) - 0x00$ $(A) - 0x00$ $(X) - 0x00$ $(M) - 0x00$ $(M) - 0x00$ $(M) - 0x00$	0	—	—	↑	↑	—	DIR INH INH IX1 IX SP1	3D 4D 5D 6D 7D 9E6D	dd ff ff	4 1 1 4 3 5
TSX	Transfer SP to Index Reg.	$H:X \leftarrow (SP) + 0x0001$	—	—	—	—	—	—	INH	95		2
TXA	Transfer X (Index Reg. Low) to Accumulator	$A \leftarrow (X)$	—	—	—	—	—	—	INH	9F		1
TXS	Transfer Index Reg. to SP	$SP \leftarrow (H:X) - 0x0001$	—	—	—	—	—	—	INH	94		2
WAIT	Enable Interrupts; Wait for Interrupt	$I \text{ bit} \leftarrow 0$; Halt CPU	—	—	0	—	—	—	INH	8F		2+

¹ Bus clock frequency is one-half of the CPU clock frequency.

Table 8-3. Opcode Map (Sheet 1 of 2)

Bit-Manipulation			Branch		Read-Modify-Write								Control				Register/Memory														
00	5	10	5	20	3	30	5	40	1	50	5	60	5	70	4	80	9	90	3	A0	2	B0	3	C0	4	D0	4	E0	3	F0	3
BRSET0	DIR	BSET0	DIR	BRA	REL	NEG	DIR	NEGA	INH	NEGX	INH	NEG	IX1	NEG	IX	RTI	INH	BGE	REL	SUB	IMM	SUB	DIR	SUB	EXT	SUB	IX2	SUB	IX1	SUB	IX
01	5	11	5	21	3	31	5	41	4	51	4	61	5	71	5	81	6	91	3	A1	2	B1	3	C1	4	D1	4	E1	3	F1	3
BRCLR0	DIR	BCLR0	DIR	BRN	REL	CBEQ	DIR	CBEQA	IMM	CBEQX	IMM	CBEQ	IX1+	CBEQ	IX+	RTS	INH	BLT	REL	CMP	IMM	CMP	DIR	CMP	EXT	CMP	IX2	CMP	IX1	CMP	IX
02	5	12	5	22	3	32	5	42	5	52	6	62	1	72	1	82	5+	92	3	A2	2	B2	3	C2	4	D2	4	E2	3	F2	3
BRSET1	DIR	BSET1	DIR	BHI	REL	LDHX	EXT	MUL	INH	DIV	INH	NSA	INH	DAA	INH	BGND	INH	BGT	REL	SBC	IMM	SBC	DIR	SBC	EXT	SBC	IX2	SBC	IX1	SBC	IX
03	5	13	5	23	3	33	5	43	1	53	1	63	5	73	4	83	11	93	3	A3	2	B3	3	C3	4	D3	4	E3	3	F3	3
BRCLR1	DIR	BCLR1	DIR	BLS	REL	COM	DIR	COMA	INH	COMX	INH	COM	IX1	COM	IX	SWI	INH	BLE	REL	CPX	IMM	CPX	DIR	CPX	EXT	CPX	IX2	CPX	IX1	CPX	IX
04	5	14	5	24	3	34	5	44	1	54	1	64	5	74	4	84	1	94	2	A4	2	B4	3	C4	4	D4	4	E4	3	F4	3
BRSET2	DIR	BSET2	DIR	BCC	REL	LSR	DIR	LSRA	INH	LSRX	INH	LSR	IX1	LSR	IX	TAP	INH	TXS	INH	AND	IMM	AND	DIR	AND	EXT	AND	IX2	AND	IX1	AND	IX
05	5	15	5	25	3	35	4	45	3	55	4	65	3	75	5	85	1	95	2	A5	2	B5	3	C5	4	D5	4	E5	3	F5	3
BRCLR2	DIR	BCLR2	DIR	BCS	REL	STHX	DIR	LDHX	IMM	LDHX	DIR	CPHX	IMM	CPHX	DIR	TPA	INH	TSX	INH	BIT	IMM	BIT	DIR	BIT	EXT	BIT	IX2	BIT	IX1	BIT	IX
06	5	16	5	26	3	36	5	46	1	56	1	66	5	76	4	86	3	96	5	A6	2	B6	3	C6	4	D6	4	E6	3	F6	3
BRSET3	DIR	BSET3	DIR	BNE	REL	ROR	DIR	RORA	INH	RORX	INH	ROR	IX1	ROR	IX	PULA	INH	STHX	EXT	LDA	IMM	LDA	DIR	LDA	EXT	LDA	IX2	LDA	IX1	LDA	IX
07	5	17	5	27	3	37	5	47	1	57	1	67	5	77	4	87	2	97	1	A7	2	B7	3	C7	4	D7	4	E7	3	F7	2
BRCLR3	DIR	BCLR3	DIR	BEQ	REL	ASR	DIR	ASRA	INH	ASRX	INH	ASR	IX1	ASR	IX	PSHA	INH	TAX	INH	AIS	IMM	STA	DIR	STA	EXT	STA	IX2	STA	IX1	STA	IX
08	5	18	5	28	3	38	5	48	1	58	1	68	5	78	4	88	3	98	1	A8	2	B8	3	C8	4	D8	4	E8	3	F8	3
BRSET4	DIR	BSET4	DIR	BHCC	REL	LSL	DIR	LSLA	INH	LSLX	INH	LSL	IX1	LSL	IX	PULX	INH	CLC	INH	EOR	IMM	EOR	DIR	EOR	EXT	EOR	IX2	EOR	IX1	EOR	IX
09	5	19	5	29	3	39	5	49	1	59	1	69	5	79	4	89	2	99	1	A9	2	B9	3	C9	4	D9	4	E9	3	F9	3
BRCLR4	DIR	BCLR4	DIR	BHCS	REL	ROL	DIR	ROLA	INH	ROLX	INH	ROL	IX1	ROL	IX	PSHX	INH	SEC	INH	ADC	IMM	ADC	DIR	ADC	EXT	ADC	IX2	ADC	IX1	ADC	IX
0A	5	1A	5	2A	3	3A	5	4A	1	5A	1	6A	5	7A	4	8A	3	9A	1	AA	2	BA	3	CA	4	DA	4	EA	3	FA	3
BRSET5	DIR	BSET5	DIR	BPL	REL	DEC	DIR	DECA	INH	DECX	INH	DEC	IX1	DEC	IX	PULH	INH	CLI	INH	ORA	IMM	ORA	DIR	ORA	EXT	ORA	IX2	ORA	IX1	ORA	IX
0B	5	1B	5	2B	3	3B	7	4B	4	5B	4	6B	7	7B	6	8B	2	9B	1	AB	2	BB	3	CB	4	DB	4	EB	3	FB	3
BRCLR5	DIR	BCLR5	DIR	BMI	REL	DBNZ	DIR	DBNZA	INH	DBNZX	INH	DBNZ	IX1	DBNZ	IX	PSHH	INH	SEI	INH	ADD	IMM	ADD	DIR	ADD	EXT	ADD	IX2	ADD	IX1	ADD	IX
0C	5	1C	5	2C	3	3C	5	4C	1	5C	1	6C	5	7C	4	8C	1	9C	1	AC	8	BC	3	CC	4	DC	4	EC	3	FC	3
BRSET6	DIR	BSET6	DIR	BMC	REL	INC	DIR	INCA	INH	INCX	INH	INC	IX1	INC	IX	CLRH	INH	RSP	INH	CALL	EXT	JMP	DIR	JMP	EXT	JMP	IX2	JMP	IX1	JMP	IX
0D	5	1D	5	2D	3	3D	4	4D	1	5D	1	6D	4	7D	3	8D	7	9D	1	AD	5	BD	5	CD	6	DD	6	ED	5	FD	5
BRCLR6	DIR	BCLR6	DIR	BMS	REL	TST	DIR	TSTA	INH	TSTX	INH	TST	IX1	TST	IX	RTC	INH	NOP	INH	BSR	REL	JSR	DIR	JSR	EXT	JSR	IX2	JSR	IX1	JSR	IX
0E	5	1E	5	2E	3	3E	6	4E	5	5E	5	6E	4	7E	5	8E	2+	9E	Page 2	AE	2	BE	3	CE	4	DE	4	EE	3	FE	3
BRSET7	DIR	BSET7	DIR	BIL	REL	CPHX	EXT	MOV	DD	MOV	DIX+	MOV	IMD	MOV	IX+D	STOP	INH			LDX	IMM	LDX	DIR	LDX	EXT	LDX	IX2	LDX	IX1	LDX	IX
0F	5	1F	5	2F	3	3F	5	4F	1	5F	1	6F	5	7F	4	8F	2+	9F	1	AF	2	BF	3	CF	4	DF	4	EF	3	FF	2
BRCLR7	DIR	BCLR7	DIR	BIH	REL	CLR	DIR	CLRA	INH	CLR	INH	CLR	IX1	CLR	IX	WAIT	INH	TXA	INH	AIX	IMM	STX	DIR	STX	EXT	STX	IX2	STX	IX1	STX	IX

INH Inherent
 IMM Immediate
 DIR Direct
 EXT Extended
 DD DIR to DIR
 IX+D IX+ to DIR
 REL Relative
 IX Indexed, No Offset
 IX1 Indexed, 8-Bit Offset
 IX2 Indexed, 16-Bit Offset
 IMM to DIR
 DIR to IX+
 SP1 Stack Pointer, 8-Bit Offset
 SP2 Stack Pointer, 16-Bit Offset
 IX+ Indexed, No Offset with Post Increment
 IX1+ Indexed, 1-Byte Offset with Post Increment

Opcode in Hexadecimal
 Number of Bytes
 F0 SUB 3
 1 IX
 HCS08 Cycles
 Instruction Mnemonic
 Addressing Mode

Table 8-3. Opcode Map (Sheet 2 of 2)

Bit-Manipulation	Branch	Read-Modify-Write				Control				Register/Memory							
						9E60 6 3 SP1 NEG						9ED0 5 4 SP2 SUB	9EE0 4 3 SP1 SUB				
						9E61 6 4 SP1 CBEQ						9ED1 5 4 SP2 CMP	9EE1 4 3 SP1 CMP				
												9ED2 5 4 SP2 SBC	9EE2 4 3 SP1 SBC				
						9E63 6 3 SP1 COM						9ED3 5 4 SP2 CPX	9EE3 4 3 SP1 CPX	9EF3 6 3 SP1 CPHX			
						9E64 6 3 SP1 LSR						9ED4 5 4 SP2 AND	9EE4 4 3 SP1 AND				
												9ED5 5 4 SP2 BIT	9EE5 4 3 SP1 BIT				
						9E66 6 3 SP1 ROR						9ED6 5 4 SP2 LDA	9EE6 4 3 SP1 LDA				
						9E67 6 3 SP1 ASR						9ED7 5 4 SP2 STA	9EE7 4 3 SP1 STA				
						9E68 6 3 SP1 LSL						9ED8 5 4 SP2 EOR	9EE8 4 3 SP1 EOR				
						9E69 6 3 SP1 ROL						9ED9 5 4 SP2 ADC	9EE9 4 3 SP1 ADC				
						9E6A 6 3 SP1 DEC						9EDA 5 4 SP2 ORA	9EEA 4 3 SP1 ORA				
						9E6B 8 4 SP1 DBNZ						9EDB 5 4 SP2 ADD	9EEB 4 3 SP1 ADD				
						9E6C 6 3 SP1 INC											
						9E6D 5 3 SP1 TST											
										9EAE 5 2 IX LDHX	9EBE 6 4 IX2 LDHX	9ECE 5 3 IX1 LDHX	9EDE 5 4 SP2 LDX	9EEE 4 3 SP1 LDX	9EFE 5 3 SP1 LDHX		
						9E6F 6 3 SP1 CLR						9EDF 5 4 SP2 STX	9EEF 4 3 SP1 STX	9EFF 5 3 SP1 STHX			

INH Inherent REL Relative SP1 Stack Pointer, 8-Bit Offset
 IMM Immediate IX Indexed, No Offset SP2 Stack Pointer, 16-Bit Offset
 DIR Direct IX1 Indexed, 8-Bit Offset IX+ Indexed, No Offset with
 EXT Extended IX2 Indexed, 16-Bit Offset Post Increment
 DD DIR to DIR IMD IMM to DIR IX1+ Indexed, 1-Byte Offset with
 IX+D IX+ to DIR DIX+ DIR to IX+ Post Increment

Note: All Sheet 2 Opcodes are Preceded by the Page 2 Prebyte (9E)

Prebyte (9E) and Opcode in
 Hexadecimal 9E60 6
 3 SP1
 Number of Bytes 3 NEG HCS08 Cycles
 SP1 Instruction Mnemonic
 Addressing Mode

